

Question 1

```
# Function to calculate and display square
def square_number(num):
    print("Square =", num ** 2)

# Main program
while True:
    num = int(input("Enter a number: "))

    if num < 0:
        print("Negative number entered. Program terminated.")
        break

    square_number(num)
```

```
# Open the file in read mode
file = open("data.txt", "r")

# Read and print the file line by line
for line in file:
    print(line, end="")

# Close the file
file.close()
```

Question 2

```
rainfall = [50, 80, 60, 30, 120, 90, 70, 100, 85, 95, 40, 65]  
print(rainfall)
```

b) Output of given code

</> Python

120

30

c) Errors and corrected program

Errors:

average is not defined. The variable name used is avg.

Correct code:

</> Python

```
rainfall = [50, 80, 60, 30, 120, 90, 70, 100, 85, 95, 40, 65]
```

```
avg = sum(rainfall) / 12
```

```
print("average rainfall =", avg)
```

d) Print months with dry rainfall

Assume rainfall less than 70 mm is dry.

</> Python

```
rainfall = [50, 80, 60, 30, 120, 90, 70, 100, 85, 95, 40, 65]

months = ["January", "February", "March", "April", "May", "June",
          "July", "August", "September", "October", "November", "December"]

count = 0

for i in range(12):
    if rainfall[i] < 70:
        print(months[i], "has Dry rainfall:", rainfall[i], "mm")
        count = count + 1

print("Total months with Dry rainfall:", count)
```

Output:

</> Python

```
January has Dry rainfall: 50 mm
March has Dry rainfall: 60 mm
April has Dry rainfall: 30 mm
November has Dry rainfall: 40 mm
December has Dry rainfall: 65 mm
Total months with Dry rainfall: 5
```

Q No. 3 Solution

(i) Program to Check Attendance of One Student

Python

```
status = input("Enter attendance (present/absent): ")

if status == "present":
    print("Student is present.")
elif status == "absent":
    print("Student is absent.")
else:
    print("Invalid input.")
```

```

present_count = 0

for i in range(1, 6):
    status = input("Enter attendance of Student " + str(i) + " (present/absent): ")

    if status == "present":
        print("Student is present.")
        present_count = present_count + 1

    elif status == "absent":
        print("Student is absent.")

    else:
        print("Invalid input.")

print("Total students present =", present_count)

percentage = (present_count / 5) * 100

print("Attendance Percentage =", percentage, "%")

```

(a) Algorithm to Input Three Numbers and Print the Smallest Number (3 Marks)

Algorithm

1. Start
2. Input three numbers A, B, and C
3. If $A \leq B$ and $A \leq C$, print A as the smallest number
4. Else if $B \leq A$ and $B \leq C$, print B as the smallest number
5. Else print C as the smallest number
6. Stop

Pseudocode

```
START
INPUT A, B, C

IF A <= B AND A <= C THEN
    PRINT A
ELSE IF B <= A AND B <= C THEN
    PRINT B
ELSE
    PRINT C
END IF

STOP
```

(b) Binary Search Algorithm to Find 36 (3 Marks)

Step 1: Binary Search requires a sorted list

The given list is:

93, 87, 83, 71, 65, 52, 47, 36, 10

Sort it in ascending order:

10, 36, 47, 52, 65, 71, 83, 87, 93

Step 2: Apply Binary Search

Step	Low	High	Mid	Value at Mid	Result
1	0	8	4	65	36 < 65, search left half
2	0	3	1	36	Found

Working

Sorted List:

10	36	47	52	65	71	83	87	93
0	1	2	3	4	5	6	7	8

Step 1:

Low = 0, High = 8

Mid = $(0 + 8) // 2 = 4$

Value = 65

$36 < 65$

Move to left half

Step 2:

Low = 0, High = 3

Mid = $(0 + 3) // 2 = 1$

Value = 36

Found!

START

LIST = [10, 36, 47, 52, 65, 71, 83, 87, 93]

TARGET = 36

LOW = 0

HIGH = 8

WHILE LOW <= HIGH

 MID = (LOW + HIGH) // 2

 IF LIST[MID] = TARGET THEN

 PRINT "Found"

 STOP

 ELSE IF TARGET < LIST[MID] THEN

 HIGH = MID - 1

 ELSE

 LOW = MID + 1

 END IF

END WHILE

PRINT "Not Found"

STOP

Python Program to Generate a Dataset for $y = 3x + 4$ and Plot a Line Chart and Box Plot

<> Python



Run

```
import matplotlib.pyplot as plt

# Generate dataset
x = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
y = []

for i in x:
    y.append(3 * i + 4)

# Display dataset
print("x =", x)
print("y =", y)
```

Line Chart

```
plt.figure(figsize=(6, 4))
plt.plot(x, y, marker='o')
plt.title("Line Chart of  $y = 3x + 4$ ")
plt.xlabel("x")
plt.ylabel("y")
plt.grid(True)
plt.show()
```

Box Plot

```
plt.figure(figsize=(4, 4))
plt.boxplot(y)
plt.title("Box Plot of y Values")
plt.ylabel("y")
plt.show()
```

Generated Dataset

x	$y = 3x + 4$
1	7
2	10
3	13
4	16
5	19
6	22
7	25
8	28
9	31
10	34

Expected Output

Line Chart:

A straight line passing through the points:



Since the slope is 3, the graph rises uniformly as x increases.

Box Plot:

A box plot showing the distribution of the y-values:

7, 10, 13, 16, 19, 22, 25, 28, 31, 34

This plot summarizes:

- Minimum = 7
- Maximum = 34
- Median = 20.5
- Quartiles (Q1 and Q3)
- Spread of the data

This solution satisfies the requirements of:

1. Generating a dataset with two variables (x and y).
2. Using the equation $y = 3x + 4$.
3. Plotting a **line chart**.
4. Plotting a **box plot**.